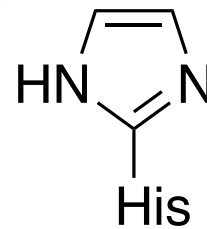
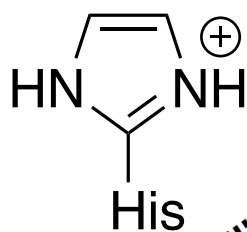


rate

$$k_{obs} = k_{A^-} \cdot \frac{K_a}{K_a + [H^+]}$$



add $\oplus$ charge

add $\ominus$ charge

pH